

Chapter 161

Resolve FX OpenColorIO

OpenColorIO (OCIO) is an open color management system created by the Academy Software Foundation. It is used by many professional vfx and editing programs (including DaVinci Resolve and Fusion) to ensure color accuracy throughout the production pipeline.

These are technical FX to integrate OpenColorIO data into the DaVinci Resolve workflow.

Contents

OCIO CDL Transform	3522
OCIO Color Space.....	3522
OCIO Display	3523
OCIO File Transform	3523

OCIO CDL Transform

The OCIO CDL Transform FX lets you create, save and apply standard Color Decision List (CDL) grades to or from the node. A CDL exchanges basic primary color information from one application to another.

- **CDL File:** Clicking Load will allow you to load a Color Decision List (.cc or .ccc) file to the controls. CC is a single CDL and CCC is an archive of multiple CDL looks. If loading a CCC file, another CCID field appears and lets you choose the name or number of the index. Type the name or number and press Reload to load the new version.
- **Reload:** Lets you reload a CDL after choosing another name or number from the index of a CCC file.
- **Export:** Exports the current settings in Controls as a new .cc file.
- **CDL Controls:** These controls let you modify the basic color information of the node that can be exported as a CDL file for other applications to read. These controls, Slope, Offset, Power and Saturation, are how the American Society of Cinematographers encode color information. The color controls are self explanatory, but without getting deep into the math, these controls allow you to adjust color in either a film or video based manner as below:
 - Slope is essentially Gain
 - Offset is essentially Exposure
 - Power is Essentially Gamma
 - Contrast is a combination of Slope and Offset
 - Lift is a combination of Slope and Offset
- **Forward:** Applies the CDL changes to the node.
- **Reverse:** Attempts to remove the CDL changes from the node. For example, if you receive some footage with a baked in color grade, reversing the CDL used will remove those changes back to the original ungraded source material. However, not all color correction can be undone.
- **Export:** Exports the current settings in Controls as a new .cc file.

OCIO Color Space

This effect allows you to do color space transforms using an OCIO config file (.ocio).

- **OCIO Config File:** Click Browse to load a .ocio config file for the color space transform.
- **Source Space:** Based on the config file, the available source color spaces are listed here. The content of this list is based solely on the loaded profile and hence can vary immensely.
- **Output Space:** Based on the config file, the available output color spaces are listed here. The content of this list is based solely on the loaded profile and hence can vary immensely.
- **Look:** Installed OCIO Color Transform Looks appear in this menu. If no looks are installed, this menu has only None listed as an option.
- **Forward:** Applies the Color Space changes to the node.
- **Reverse:** Attempts to remove the Color Space changes from the node.

OCIO Display

This effect allows you to view display transforms using an OCIO config file (.ocio).

- **OCIO Config File:** Click Browse to load a .ocio config file for the display transform.
- **Source Space:** Choose the color space of the source material from the list.
- **Display:** Choose which monitor and type your Viewer is on.
- **View:** Choose which color space to view on your monitor.

OCIO File Transform

This effect allows you to load and apply Look Up Tables (LUTs) to the node.

- **LUT File:** Click Browse to load a LUT from the file browser.
- **CCC ID:** This is the ID key used to identify the specific file transform located within the ASC CDL color correction XML file.
- **Forward:** Applies the LUT to the node.
- **Reverse:** Attempts to remove the LUT from the node.
- **Interpolation:** Allows the user to select the color interpolation to achieve the best quality/render time ratio.